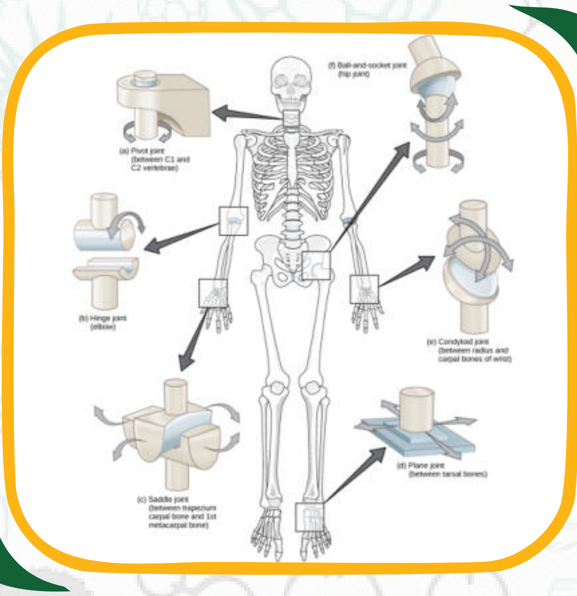
UNIT NO.4

SUPPORT AND MOVEMENT



SUPPORT AND MOVEMENT

The musculoskeletal system is a remarkable framework that provides both support and facilitates movement in the human body. Comprising bones, muscles, joints, and connective tissues, this system not only offers structural stability but also allows for a wide range of motions. From the rigid support of the skeleton to the dynamic contraction of muscles, the coordination of these elements ensures essential functions such as walking, grasping, and maintaining posture. Understanding the intricacies of support and movement is vital for comprehending the body's capabilities and addressing issues related to mobility and stability.

Types of movement based on stimuli:

TYPE OF MOVEMENT	DESCRIPTION	EXAMPLE
Autonomic Movements	Involuntary movements, not under conscious control	Heartbeat, peristalsis in the digestive system
Paratonic Movements	Involuntary movements in response to external stimuli	Reflexes, such as withdrawing hand from a hot surface

Types of movement based on responses:

TYPE OF MOVEMENT	DESCRIPTION	EXAMPLE
Tropic Movement	Plant movement in response to a directional stimulus	Phototropism (towards light)
Nastic Movement	Non-directional plant movement independent of the direction of the stimulus	Mimosa plant folding leaves when touched
Locomotory Movement	Movement involving a change in position or location	Walking, running

Difference between movement and locomotion:

Movement refers to any change in the position or posture of an organism or its parts. It includes activities like growth, bending, or turning.Locomotion, on the other hand, specifically refers to the self-powered, directed movement of an organism from one place to another, often involving the use of limbs or other specialized structures.In essence, all locomotion is a form of movement, but not all movement is locomotion.

SKELETON AND ITS TYPES

The skeleton is the internal framework of an organism, providing support and structure. Its three main functions are as follows:

• Support: Maintains the body's shape.
• Protection: Shields vital organs.
• Movement: Serves as an attachment point for muscles, facilitating motion.

Types of Skeleton:
There are three types of skeleton:

Type of Skeleton	Characteristics	Examples	Additional Point
Endoskeleton	Internal framework made of bones or cartilage.	Humans, vertebrates	Provides structure for muscle attachment.
Exoskeleton	External hard covering, provides support & protection.	Insects, crustaceans	Molting is necessary for growth.
Hydrostatic Skeleton	Fluid-filled cavity, muscles work against it for movement.	Earthworms, jellyfish	Enables soft-bodied animals to maintain shape.

Human Skeleton:
The skeletal system of human is basically made up of two types of skeleton:
- Cartilage
- Bones

Characteristic	Cartilage	Bones
Composition	Flexible, rubbery connective tissue.	Hard, rigid connective tissue.
Strength	Less dense and not as strong as bones.	Dense and provides structural strength.
Function	Cushions joints, supports nose and ears.	Provides support, protects organs.
Growth	Can grow throughout life, limited growth.	Growth occurs primarily during youth.

Skeleton provides support and movement:
The skeletal system, consisting of bones, cartilage, and joints, provides a framework that supports and shapes the body. It acts as a scaffold for muscles to attach, facilitating movement and enabling activities such as walking, running, and lifting. Additionally, the skeleton protects vital organs, such as the brain and heart, ensuring their safety within the body.

Skeleton system is a dynamic system:
The skeletal system is a dynamic and constantly changing system that undergoes processes like bone remodeling and growth throughout life. It adapts to external stimuli and mechanical stresses by modifying its structure and density. This dynamic nature allows the skeleton to respond to the body's needs, ensuring optimal support, protection, and mobility.

Model of bone healing occurs in 6 weeks:

- **Hematoma Formation:**

After a bone fracture, blood vessels are disrupted, leading to bleeding and the formation of a blood clot (hematoma) at the fracture site.

- **Fibrocartilaginous Callus Formation:**

Fibroblasts invade the hematoma and deposit collagen fibers, creating a fibrocartilaginous callus that stabilizes the fracture site.

- **Bony Callus Formation:**

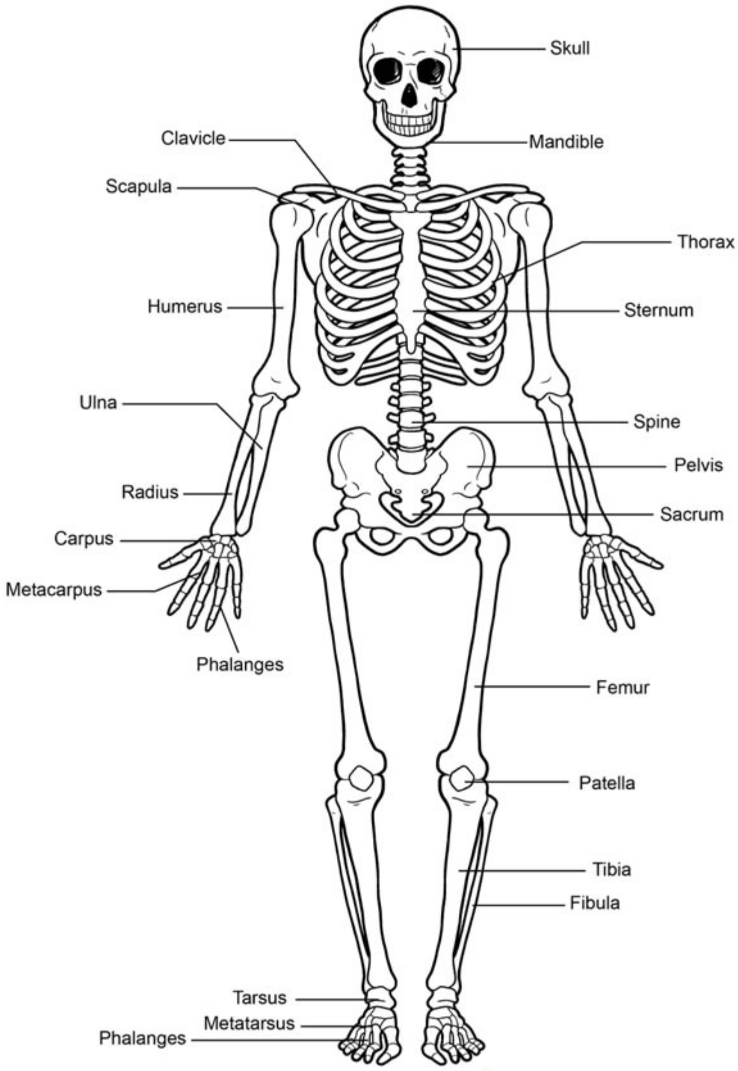
Osteoblasts then produce spongy bone trabeculae within the fibrocartilaginous callus, converting it into a bony callus. This stage restores some structural integrity.

- **Bone Remodeling:**

Over weeks to months, the bony callus undergoes remodeling, as osteoclasts remove excess bone tissue, and osteoblasts build new, compact bone. The final structure closely resembles the original bone.This four-step process typically occurs over a span of about six weeks during the bone healing process.

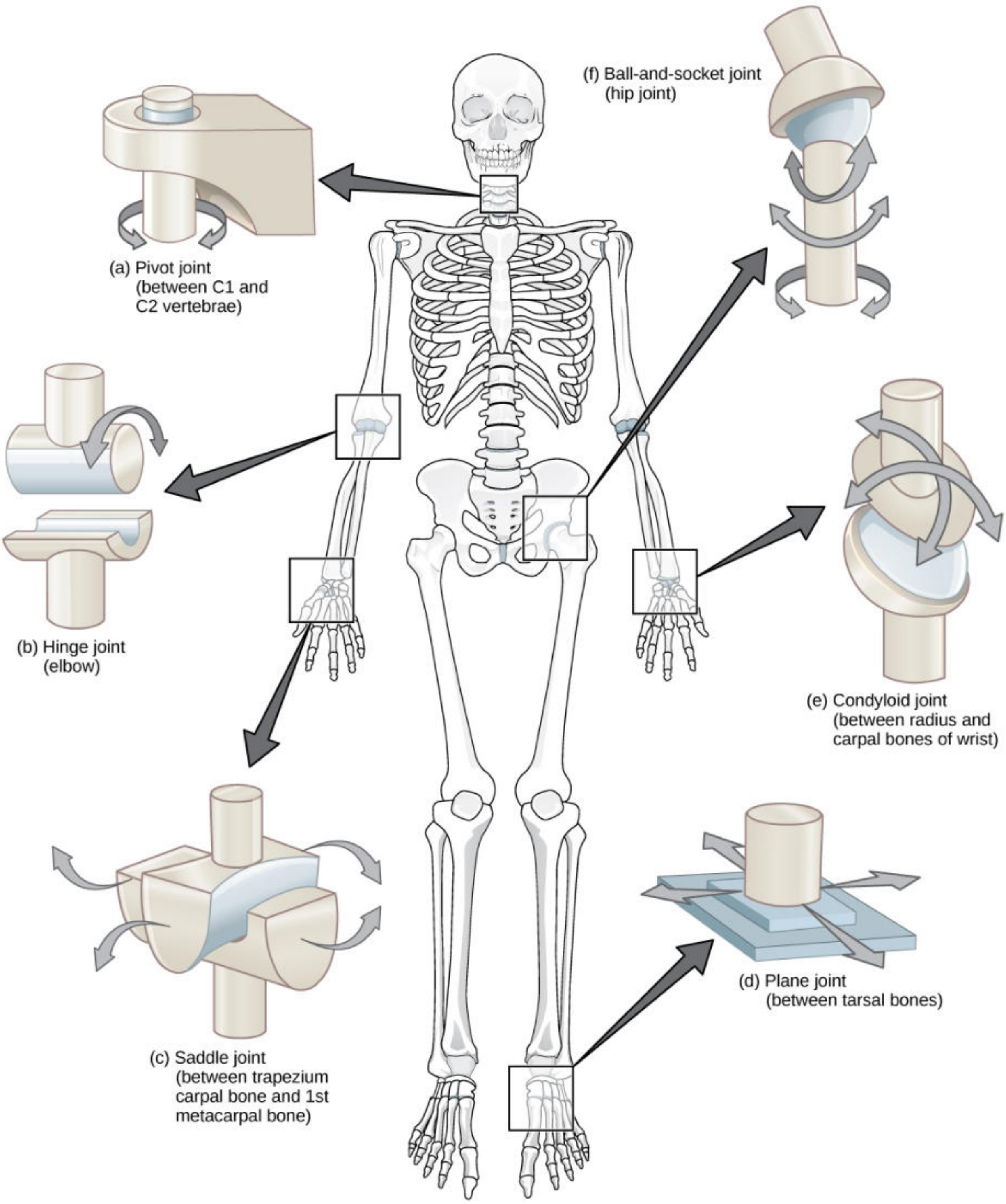
HUMAN SKELETON

The human skeleton is the internal framework of the body, providing support, protection, and attachment points for muscles. Composed of 206 bones, it is categorized into two main divisions: the axial skeleton, including the skull, vertebral column, and rib cage; and the appendicular skeleton, consisting of the limbs and girdles. Bones serve various functions, such as facilitating movement, storing minerals, and producing blood cells in the bone marrow. Joints, where bones meet, allow flexibility, and the skeletal system plays a crucial role in maintaining the body's shape and protecting vital organs.



JOINTS

Joints, also known as articulations, are crucial components of the skeletal system, connecting bones and enabling movement. They can be classified into three main types: fibrous joints, which allow minimal movement and are connected by fibrous tissue; cartilaginous joints, where bones are joined by cartilage, providing some flexibility; and synovial joints, the most common and movable type, enclosed by a joint capsule filled with synovial fluid.Synovial joints, found in the knees, shoulders, and hips, offer a wide range of motion. They consist of articular cartilage, synovial membrane, synovial fluid, ligaments, and tendons. Ligaments connect bones, stabilizing joints, while tendons attach muscles to bones, facilitating movement.The function of joints is to provide support, absorb shock, and permit smooth, controlled motion. However, issues such as arthritis can impact joint health, causing pain and reduced mobility. Proper care, exercise, and maintaining a healthy lifestyle contribute to the overall well-being of joints.



Ligament and tendon: their role in movement

Ligaments are strong, flexible connective tissues that connect bones to other bones, providing stability to joints. They play a crucial role in limiting excessive movement and preventing dislocation.Tendons are tough connective tissues that attach muscles to bones, transmitting the force generated by muscle contraction to produce movement. They are essential for joint mobility and overall body locomotion.

Location and movement of hinge joint:

Located in the elbow, a hinge joint allows movement primarily in one direction, resembling the action of opening and closing a hinge. It facilitates flexion and extension movements.

Location and movement of Ball and Socket Joint:

Found in the hip and shoulder, ball and socket joints permit a wide range of motion in multiple directions, including rotation, flexion, extension, adduction, and abduction.

MUSCLES

Muscles are contractile tissues that enable body movement by generating force through contraction. They play a vital role in various physiological functions.

Skeletal Muscles:

Skeletal muscles are attached to bones by tendons, facilitating voluntary movements like walking and reaching. They are under conscious control and form the majority of the body's muscle mass.

Cardiac Muscles:

Cardiac muscles are found in the heart, pumping blood throughout the circulatory system. They exhibit involuntary contractions, ensuring continuous and rhythmic cardiac activity.

Smooth Muscles:

Smooth muscles are located in organs like the digestive tract and blood vessels. They perform involuntary functions such as peristalsis and blood vessel constriction, contributing to organ and system regulation.

Antagonistic Muscles (Biceps and Triceps):

In the context of muscles, antagonistic refers to pairs of muscles that work in opposition to achieve controlled movement. The biceps and triceps exemplify this concept in the arm. The biceps, located in the front, contract to flex the arm (e.g., lifting a weight), while the triceps, positioned at the back, contract to extend the arm. This antagonistic interaction enables coordinated and controlled arm movements.

DISORDER OD SKELETON SYSTEM

Effect of Calcium Deficiency on Bones:

- Calcium deficiency can lead to weakened bones and an increased risk of fractures.
- Insufficient calcium intake may compromise bone density and mineralization.
- Bones serve as a reservoir for calcium; deficiency may prompt the body to draw calcium from bones.
- Prolonged deficiency can contribute to conditions like osteoporosis, characterized by brittle and porous bones.

Ricket:

Rickets is a disorder characterized by soft and weakened bones, primarily caused by vitamin D deficiency. Inadequate vitamin D impairs calcium absorption, leading to skeletal deformities and bone abnormalities in children.

Arthritis:

Arthritis is inflammation of the joints.

Symptoms:
- Pain
- Swelling
- Stiffness
- Redness
- Decreased range of motion

Common types of arthritis

Osteoarthritis:

Osteoarthritis is the degeneration of joint cartilage and the underlying bone, causing pain and stiffness.

Rheumatoid Arthritis:

Rheumatoid arthritis is an autoimmune disorder that primarily affects the joints, causing inflammation and joint deformity.

Psoriatic Arthritis:

Psoriatic arthritis is a type of arthritis that affects some people with psoriasis, causing joint pain, stiffness, and swelling.

Causes of Arthritis:
- Inflammation
- Genetics
- Age
- Injury
- Infection

Role of obesity in arthritis:

1. Obesity puts extra stress on joints, increasing the risk of arthritis.
2. Adipose tissue produces inflammatory chemicals that can worsen arthritis.
3. Obesity is a major factor in the development of osteoarthritis.
4. Losing weight can alleviate symptoms and slow the progression of arthritis in obese individuals.